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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MAR 522 — Past Questions & Materials

by Renance

Course: MAR 522

Institution: Federal University of Petroleum Resources, Effurun

Papers in this pack: 1

Levels: 500

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Paper 1 of 1 — MAR 522 SHIP PROPULSION SYSTEM SECOND SEMESTER 2022/2023

Level 500 • Second Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF MARINE ENGINEERING

Lx1

EXAMINATION: Second Semester

SESSION: 2022/2023

DATE: 11/10/2023



COURSE CODE: MAR 522

COURSE TITLE: Ship Propulsion System

TIME ALLOWED: 2 Hrs

INSTRUCTION: ATTEMPT ANY 4 QUESTIONS

Question 1

- i. Discuss the evolution of Ship Propulsion System (8.5 marks)
- ii. A ship of 5,000 Tons displacement in sea water has speed of 18 knots, length of 120 meters. A towing tank model has 6 meters length. At which speed should the model be tested. Also, what is the ratio of the ship to the model power at this speed? (9 marks)

Question 2

- i. Prove, using dimensional analysis, that the towing-tank total resistance coefficient C_T of a ship is a function of Reynolds number, Froude number and pressure coefficient: (8.5 marks)
- ii. Describe the parts of a Screw Propeller Geometry (9 marks)

Question 3

- i. In Floating structure dynamics, show that the Motion Amplitude of a structure is expressed as

$$X = \frac{F_0}{(m + A_{(\omega)})(\omega t)^2 + c_{(\omega)}X(\omega t) + K_{(\omega)}}$$

- Where the symbols have their meanings (5 marks)
- ii. A 180m long container ship with a displacement of 34,260 m³ travels at 20 knots in seawater of viscosity = 1.18831×10^{-6} m²/s against a wind of relative velocity of 20 knots.
Given that the wetted surface is 6.8723; the projected area AP and the air coefficient are 278.652 and 0.734 respectively. Wave resistance is 30% of total resistance. Calculate the effective power of the ship. (12.5 marks)

Question 4

- i. Show that for a propulsion system, the propulsive efficiency is expressed as

$$\eta_P = \eta_D \times \eta_S$$

- Where the symbols have their meanings (5 marks)
- ii. A 6m model of a 180m long ship is towed in a model basin at a speed of 2.5 m/s. The towing pull is 20N. The wetted surface of the model is 4m². Estimate the corresponding speed for the ship in knots and the total resistance using simple scaling laws. From the resistance calculate the effective power PE. (12.5 marks)